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Freely typed keystroke dynamics-based user authentication for mobile devices based on heterogeneous features

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ABSTRACT

Keystroke dynamics-based authentication (KDA) is one of the human behavioral biometric-based user authentication methods based on the unique typing pattern of a person. Previous KDA studies on mobile devices primarily focused on fixed-length text-based KDA, such as passwords and personal identification numbers. This can strengthen the login system and prevent abnormal usage of impostors based on certain attack methods, such as shoulder surfing and smudge attacks. However, this method possesses a limitation that continuous monitoring is not possible after login. To solve this problem, KDA based on freely typed text was studied; however, there are only a few studies on this technique. Further, the performance authentication based on these studies is insufficient for a real-world implementation. In this paper, we propose a novel freely typed text-based KDA method for mobile devices named FACT, i.e., user authentication on mobile devices based on free text, accelerator, coordinate, and time. We collected data from three different smartphone sensors while typing in two languages (English and Korean), and 17 variables were extracted for a set of keystroke data. A total of six authentication methods were employed and the proposed FACT yielded an equal error rate lower than 1% with only one reference keystroke set; moreover, it demonstrated a perfect protection capability while using Korean when more than four reference keystroke sets were used. To contribute to the research and industrial community, we have publicized our collected keystroke dataset so that anyone who conducts a KDA study or develops a KDA-related mobile service can use the dataset without any restrictions.

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1. Introduction

Today, we are living in a world where network connectivity is rapidly increasing; furthermore, the development of mobile devices provides significant conveniences by recording and analyzing the behaviors of users. However, the scope of a security system is also more extensive, and security-related concerns are rapidly evolving [1,2]. Consequently, companies are investing heavily in hardware and software for security systems; however, many network systems remain vulnerable to security breaches, resulting in personal/corporate information leakage/loss. Generally, security systems consist of different levels of security coverage; the first level and the most atomic unit in the entire security system is user authentication, which focuses on discriminating potential impostors from a valid user based on the personal information of the valid user. A user authentication system grants an authorized user access to a request only when a set of required information is verified. Therefore, the simplest user authentication system in a mobile device is to compare the personal identification number (PIN) or password during a login trial [3]. The distinctive characteristics of password- and PIN-based authentication methods are the ease of development, operation, and maintenance. However, if a third party obtains the valid PIN or password, the security system becomes useless [4].

To solve this problem, a physical biometric-based user authentication system was developed [5]. Physical biometric features, such as the frontal face, profile face, fingerprints, finger photo, iris, voice, and blood vessels, do not require users to carry a separate security token such as a one-time PIN or password. In addition, these physical biometric-based user-authentication methods have achieved satisfactory authentication performances in several studies [6–10]. However, face recognition, fingerprint, and iris-based authentication methods may also be inefficient when used on an image with a low resolution or different lighting conditions or on a print of a forged face, fingerprint, or iris [11–14]. In addition, physical biometric-based systems require additional hardware to read

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the physical biometric information; consequently, the implementation scope is limited to certain devices in which the recognition hardware is installed. Therefore, physical biometric-based system may become expensive and a burden on physical space.

To overcome this limitation, keystroke dynamics-based authentication (KDA), which is one of the human behavior biometric-based user authentication methods, was studied [15–18]. KDA is based on the unique typing pattern of each person. Therefore, it does not require additional hardware and it can be fully implemented using software programs. Thus, KDA demonstrates a lesser burden on expense and physical space than physical biometric-based methods. Previous studies of KDA can be broadly categorized into either fixed-length text-based KDA (fix-KDA) or freely typed text-based KDA (free-KDA). Fix-KDA is based on a fixed length of text, such as a PIN or password. The fix-KDA can strengthen the login system and prevent threats, such as shoulder surfing attack (SSA) and smudge attack (SA); however, in this method, the users cannot be authenticated or continually monitored after the login process.

To solve this problem, free-KDA was studied. Although free-KDA for personal computer (PC) keyboards was actively studied [19–22], there are relatively few studies in real-world situations using various sensor data from mobile devices. In addition, in general, free-KDA for mobile devices yielded lower user authentication performances than free-KDA for PC keyboards [23].

In this paper, we propose a novel free-KDA methodology for mobile devices, named as FACT, i.e., "freely typed keystroke dynamics micro behavior based on a mixture of accelerator, coordinate, and time for mobile user authentication. Several characteristics of FACT are described subsequently. First, it is a real-world situation-based user authentication methodology, which uses only the keystroke data of valid users for the authentication process. Second, it is based on a mixture of three microbehavioral feature set from accelerometer sensor, touch coordinate, and time sensor. Third, it yielded sufficiently high user authentication performance for real-world security systems regardless of the languages tested, i.e., English and Korean. Fourth, it can enlarge the individual user-level security scope because it is not only robust to threats, such as SSA and SA, but also available for user authentication or continual monitoring even after login. Finally, it can be fully implemented using software programs; therefore, it is less expensive than physical biometric-based methods.

As another contribution to this study, we have shared the collected data used in this study with the research society. As many researchers have difficulty acquiring real datasets, we collected mobile KDA data by carefully designing the data collection experiment. We expect that the results of this study and the shared data can help to initiate novel ideas in KDA based on freely typed text in mobile devices.

The remainder of this paper is organized as follows. In Section 2, we review previous KDA research, focusing on KDA for mobile devices. In Section 3, we introduce data collection and data preprocessing methods. In Section 4, we describe experimental design and authentication algorithms for this study, followed by a discussion on the experimental results in Section 5. Finally, we conclude the current study by discussing certain limitations and future research directions in Section 6.

2. Related work

Typical user authentication methods for mobile devices are generally based on either PIN, password, or pattern lock. These methods are inherently vulnerable to threats, such as SSA and SA, because they are authenticated based on predefined keys [24,25]. Another drawback is the decrease in convenience when a user sets a complex key combination for authentication process; consequently, users prefer simple key combinations, which can easily be stolen [26].

As an alternative to PIN, password, and pattern lock techniques, KDA, as a behavior biometric-based user authentication, was studied [15,27]. Most early KDA studies focused on a hard keyboard using physical buttons for unlocking a PC. Owing to the environmental constraints of the hard keyboard, most of these studies used only time-based features [28]. Certain recent studies attempted to use additional features, such as pressure, in addition to time-based features [29]; however, these techniques required an additional hardware to be attached to the PC keyboard to capture these features. Therefore, its expandability is limited in the real-world environment. Since the 2000s, mobile keyboard-based KDA has been studied owing to the increase in device supply ratio and mobile device portability [30,31]. However, these studies still used time-based features captured using a hard keyboard that is built in the mobile device.

Since 2010, smartphones with built-in touchscreen-based soft keyboard and various high-performance sensors are extensively utilized around the world. Aided by this technological advance, many smartphone-based KDA methods were analyzed [28]. Table 1 summarizes the previous KDA studies based on mobile devices with soft keyboards that used touch screen, from which certain findings were extracted in our study. First, smartphone-based KDA used not only conventional time (***T***)-based feature but also additional feature sets from various available sensors, such as accelerometer (***A***), gyroscope (***G***), pressure (***P***), oriented sensor (***O***) touch size (***TO***), and touch coordinate (***TC***). In contrast to PC keyboard- and hard keyboard-based mobile phones, these sensor-based features are easily expandable in smartphone-based KDA as they are already built in the smartphones. Additional features extracted from these sensors significantly improved authentication performance when they were appropriately integrated with the conventional time-based features for fix-KDA [32,33]. Second, most soft keyboard-based KDA studies focused on a predefined fixed n-digit PIN or password-based fix-KDA [34–39]. The fix-KDA was primarily studied to protect valid users from threats, such as SSA and SA, during the login process. However, these methodologies demonstrate a problem where users cannot be authenticated after the login process [40]. Conversely, the free-KDA can identify the validity of the current users during the login process as well as continually monitor the user by his/her typing behavior after login. Based on these characteristics, hard keyboard-based free-KDA was actively studied [22,41,42]; however, free-KDA based on smartphone devices was hardly studied [23,43,44].

Third, the authentication performance of free-KDA is comparatively lower than that of fix-KDA. To the best of our knowledge, the first research on free-KDA was conducted by Gunetti and Picardi [20]. Their experimental results indicated that 90% of the authentication accuracy was achieved for short and fixed texts, but only 23% of authentication accuracy was achieved for long and freely typed texts, which implies that free-KDA is significantly more difficult than fixed-KDA. Similar results were determined in mobile device-based free-KDA, as listed in Table 1.

Fourth, most mobile device-based free-KDA studies were formulated as a multiclass classification. Thus, their experimental environments were unlike a real-world situation in which only the keystroke data of valid users are available for building authentication models. Certain representative free-KDA studies based on mobile devices are as discussed subsequently. Draffin et al. [43] proposed a multi-layer perceptron-based classification model for smartphone-based free-KDA. They used ***T***-, ***TC***-, ***TO***-, ***P***- and ***G***-based features and attempted to authenticate users using 15 consecutive key presses based on the sliding window method. Their experimental results with 13 participants demonstrated a 14% false acceptance rate (FAR) and 2.2% false rejection rate (FRR). The lim-

Table 1
Related studies on KDA based on soft keyboards on touchscreens in smartphones (numbers in parentheses denote the length of data; A: accelerometer, BN: Bayesian networks, G: gyroscope, Gauss: Gaussian density estimation, *k*-NN: *k*-nearest neighbor, NB: Nave Bayesian, ND: novelty detection, MLP: multilayer perceptron, O: oriented sensor, OCSVM: one-class support vector machine, P: pressure, Parzen: Parzen window kernel density estimation, SC: supervised classification, SVDD: support vector data description, T: time, TC: touch coordinate, TO: touch size).

Study	Variable	Number of participants	Input structure	Performance	Algorithm	SC/ND	Type of Keystroke
Sougata and Kartik [34]	***T, P***	10	Digit PIN (4)	84.2% ACC	J48, NB, K*, MLP	SC	Fixed
M.V. Ignacio et al. [35].	***A, P, T, TO***	80	Digit PIN (4)	90% ACC	MLP	SC	Fixed
L. Jain et al. [36]	***P, T, TC***	30	Digit PIN (10)	2.8% EER	OCSVM	ND	Fixed
G. Ho [37]	***A, T***	55	Digit PIN (4)	FAR 0.9%, FRR 0.9%	Distance- based, RF, SVM	ND, SC	Fixed
N. Zheng et al. [32].	***A, G, P, T, TO***	80	Digit PIN (4, 8)	4-digit: 3.65% EER, 8-digit: 4.55% EER	Distance- based	ND	Fixed
M. Antal et al. [38].	***P, T, TC***	42	Password (10)	RF: 93% ACC Manhattan: 12.9% EER	Distance- based NB, BN, J48, k-NN, SVM, RF, MLP	ND, SC	Fixed
P.S. Teh et al. [33].	***P, T, TO***	150	Digit PIN (4, 16)	4-digit: 8.55% EER, 16-digit: 5.49% EER	Statistical method	ND	Fixed
H.G. Lee et al. [39]	***A, G, T, TO***	22	Digit PIN (6)	7.89% EER	Distance- based, OCSVM	ND	Fixed
B. Draffin et al. [43].	***G, P, T, TC, TO***	13	Free text based key presses (15)	FAR 14%, FRR 2.2%	MLP	SC	Free
H. Gascon et al. [44].	***A, G, O, T***	315 /12	Predefined free text (150)	TPR 92% FPR 1%	SVM	SC	Free
Kang and Cho [23]	***T***	35	Predefined free text (3000)	8.99% EER	Statistical method, k-NN, Gauss, Parzen, SVDD	ND	Free

itations of this study are its unrealistic experimental setting (formulated as a classification problem) and low authentication performance. Gascon et al. [44] gathered 315 participants, of which 12 participants were considered as valid users. Each keystroke sample was collected while typing a predefined text with an average length of 150 keystrokes. They extracted behavior biometric features from the raw keystroke data as follows. First, nine signals the combinations of three axes (x-, y-, and z-axis) and three features (***A,G*** and ***O***) were collected when the user typed the text for a specific period of time. Second, the collected nine features were transformed by three normalization methods (mean, linear, and spline). Third, they derived 88 variables and then, each variable was transformed 27 normalized variables ($9 \times 3 = 27$), which in turn resulted in $9 \times 3 \times 88 = 2,376$ variables for each keystroke set. A support vector machine was constructed using those variables for the free-KDA model. The best performance of a valid user was a true positive rate (TPR) of 92% and false positive rate (FPR) of 1%. However, the worst performance of a valid user yielded a TPR of 58% and a FPR of 35%, which is unacceptable in a practical authentication system. The limitations of this study are: (1) they formulated the authentication as a classification model; (2) they did not use key information; and (3) user authentication performances were significantly lower than keyboard-based free-KDA. Kang and Cho [23] collected raw long and freely typed keystroke data from 35 participants. Each participant typed approximately 3000 keystrokes using various devices, such as a soft keyboard in a smartphone, stylus pen, and PC keyboard. Further, they designed an authentication experiment to simulate a real-world situation where the keystroke data of only valid users were available for building authentication models. They used Gaussian estimation, Parzen window kernel estimation, *k*-nearest neighbor, support vector data description, and other statistical methods with only latency features for a free-KDA model. When users typed using two hands on a smartphone-based soft keyboard, the Cramrvon Mises criterion (CM) yielded the best performance, i.e., an equal error rate (EER) of 8.99% with 1000 keystrokes for both training and test data. However, when the length of the test keystrokes was reduced to 500, EER increased to 13.78%. Although this study reflected the real-world user authentication situation, it has certain limitations: a limited keystroke feature usage (***T***-based latency feature) and low user authentication performance.

Based on the findings and limitations observed from previous studies, the contributions of this study can be summarized as follows. First, we collected heterogeneous features, i.e., keystroke ***T***-based, ***A***-based, and ***G***-based features, when users were typing text using smartphones. Second, we proposed a new user authentication model named FACT, which integrates the above three heterogeneous features. Third, we succeeded in achieving authentication performances that can be applied to real-world systems. The proposed FACT model yielded an EER that was lower than 0.5% when only one set of keystroke data was used to train the model; moreover, it decreased to lower than 0.1% when ten sets of keystroke data were used. Finally, we publicized the collected keystroke dataset so that it is available without restrictions to any researcher who wants to conduct a study on KDA or develop KDA-related mobile services. As real data acquisition is one of the main obstacles in KDA-related studies, we expect that the developed FACT method as well as the publicized real keystroke dataset can contribute to the research and industrial communities.

3. Data collection and preprocessing

In this study, we developed an Android-based keystroke data collector, as shown in Fig. 1. Although an Android-based application (app) should work identically for different devices and versions, there are certain variations depending on the OS versions and types of devices. Therefore, to ensure the consistency of collected data, we provided the identical devices with the same OS version (Samsung Galaxy S8, Android 9.0) to the participants for the experiment. A total of 50 participants were involved in the experiment. Each participant typed 20 predefined text scripts using two hands (thumbs) in a sitting position. The average length of scripts was approximately 200 keystrokes consisting of only consonant, vowel, shift, period, and comma keys. Thus, characters with low frequency of occurrence (e.g., special characters) were excluded from the script. To investigate the effect of typing language on KDA, the participants provided their keystroke data in both English and Korean. The experimental settings, average length, and

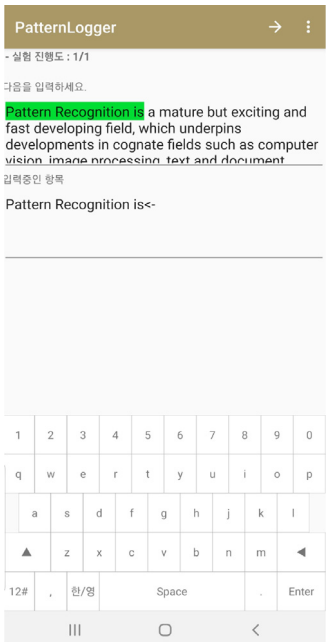


Fig. 1. Android-based keystroke data collector.

Table 2
Experimental settings and statistics of the collected keystroke data (values in parenthesis in the last two columns indicate standard deviation).

Experimental factor	Value
Total participants	50
Number of data samples	20
Language	2 (English, Korean)
Device	Samsung Galaxy S8
Average keystrokes per sample (English)	211.75 (37.88)
Average keystroke per sample (Korean)	235.19 (42.54)

Table 3
Type of keystroke features.

Name	Description
Key	Key value from keyboard
TS	Timestamp at collected data (unit : nanosecond)
Act	Status of finger at touch screen, $ACT \in \{D(\text{Down}), U(\text{Up})\}$
Cx	The coordinate of X from touched key
Cy	The coordinate of Y from touched key
Ax	Angular velocity of x-axis
Ay	Angular velocity of y-axis
Az	Angular velocity of z-axis

standard deviation of the collected keystroke data are listed in Table 2.

The user interface of the soft keyboard used in our experiment is shown in Fig. 2. The 31 keys in red color are individual keys consisting of consonants, vowels, shift, period, and backspace. The remaining keys in blue color were regarded as a single key. Thus, when a participant touched a specific key on the keyboard, it could be represented as a 32-dimensional one-hot vector.

Table 3 lists the raw keystroke features collected from the app. Key is one of the predefined 32 keys explained above. Timestamp (*TS*) is the absolute time recorded in nanosecond precision from the smartphone. Act refers to two possible activities of users: Down (*D*) for touching the key and Up (*U*) for releasing the key. When we use a hard keyboard, such as a PC keyboard for typing, we press a physical button to type a character. Conversely, when we use a soft keyboard in a smartphone for typing, we touch a key on the touchscreen instead of pressing a physical button. Hence, it is possible to record the fingertip when we touch a key in the soft keyboard. *Cx* and *Cy* refer to the coordinates of the x- and y-axes, respectively, in the key button on the touchscreen. They represent the fingertip movements while the key button is being touched and it is repeatedly recorded according to the predefined time unit of the device. Fig. 3 shows examples of the key coordinate system of a and space keys in our app and Fig. 4 shows two examples of collecting *Cx* and *Cy* for a key. In this example, Persons 1 and 2 have different *Cx* and *Cy* coordinates with different durations. All keys, except three keys (space, left shift, and backspace), have the same size. Based on our explanatory data analysis, participants showed significantly different touching habits for the same key. For example, Fig. 5 shows the scatter plots of Cx and Cy of two participants (P1 and P37) for two keys: a and space. Based on these plots, P1 habitually touches the upper-right part of the key a and the left part of the key space. P37 demonstrated the exact opposite pattern showing a tendency to touch the lower-left part of the key a and the right part of the key space. These different touching patterns might be caused by the hand size, finger length, fingertip size, muscle flexibility, and typing habit of each person, which can be useful features for user identification and authentication.

Table 4
Example of raw data1 (D and U denote Down and Up, respectively).

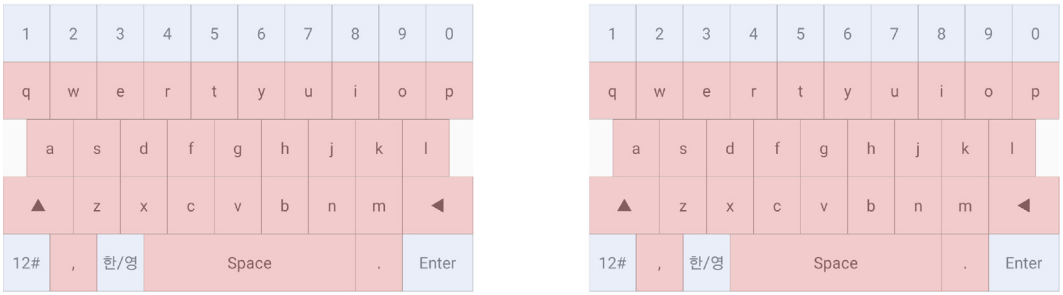
Key	*TS*(*ms*)	*Act*	*Ax*	*Ay*	*Az*
a	0	D	0.62	-0.67	-0.18
a	250	U	-0.34	0.21	0.05
b	300	D	-0.14	0.12	-0.02
b	500	U	-0.05	0.09	0.02

Table 5
Example of raw data2.

Key	*TS*(*ms*)	*Cx*	*Cy*
a	50	41.76	73.68
a	100	43.57	73.90
a	150	44.15	74.77
a	200	48.09	78.02
b	350	39.88	94.97
b	400	40.91	96.03
b	450	41.96	97.11

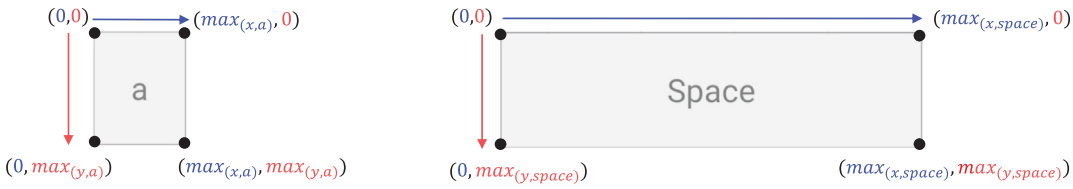
Ax, *Ay*, and *Az* refer to the angular velocities of the x-, y-, and z-axes, which are collected from the accelerometer. In this study, the angular velocities were recorded for two actions: touching and releasing a key. An example of angular velocity measurement is shown in Fig. 6. Figs. 7 and 8 show the scatterplots for two angular velocities of P1 and P37 for touching and releasing the space key, respectively. Based on Figs. 7 and 8, we can observe that the angular velocity values for P1 and P37 are significantly different for a certain combination, e.g., Ay when touching the key. These differences might be caused by different typing habits of different people, such as coordinate of fingertip on the touchscreen. To sum up, if the user types a and b consecutively, the raw data is collected as listed in Tables 4 and Table 5.

Based on the raw data listed in the above two tables, the variables defined in Table 6 are extracted as shown in Table 7. In this example, Key_1 is a and Key_2 is b. Further, the set of coordinate variables based on the first key a is represented as $Cx_1 \in \{41.76, 43.57, 44.15, 48.09\}$ and $Cy_1 \in \{73.68, 73.90, 74.77, 78.02\}$. We used the minimum ($X_{\min}$, $Y_{\min}$), average (X_{avg}, Y_{avg}), and maximum ($X_{\max}$, $Y_{\max}$) of each axis coordinate to represent the touch habit of each participant (e.g., touch position, touch drift).



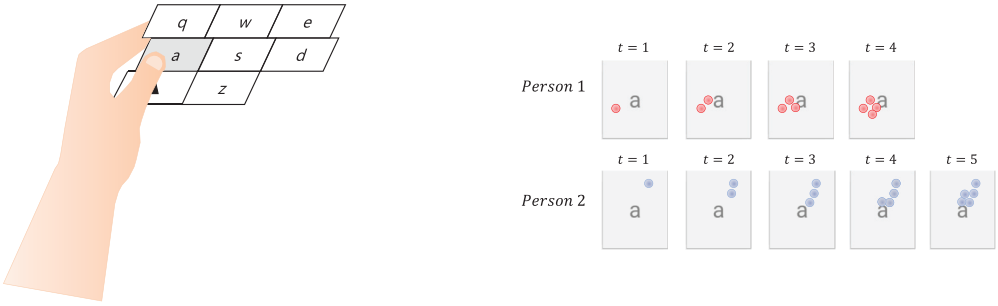
(a) English keyboard (b) Korean keyboard

Fig. 2. Keysets on the mobile keyboard used for this study.



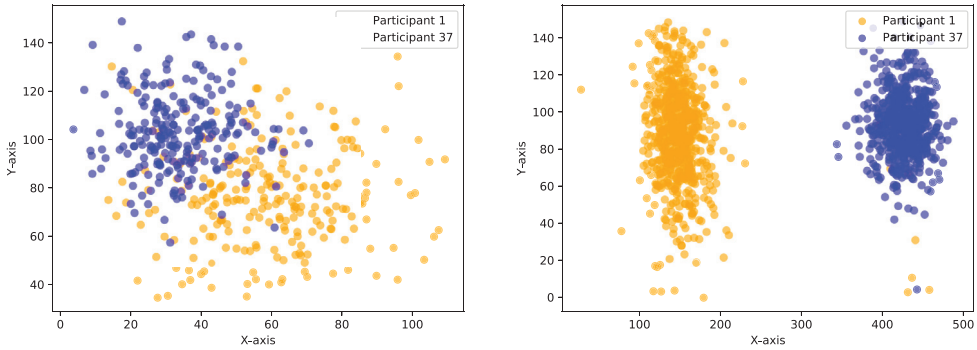
(a) Coordinate system for key "a" (b) Coordinate system for key "space"

Fig. 3. Key coordinate system in the application.



(a) Example of touching "a" key (b) Cx and Cy of "a" key collected by each person

Fig. 4. Example of Cx and Cy collection for a key (t is the minimum time collection unit of the smartphone).



(a) "a" key (b) "space" key

Fig. 5. Scatter plots of larger Cx and Cy of two participants (P1 and P37) for two keys: a and space.

Based on the transformed data, we extracted five types of keystroke timing values: duration (unigraph), down-down (DD)-time, up-up (UU)-time, up-down (UD)-time, and down-up (DU)-time (four digraphs). Duration is the time between touching and releasing a single key. DD-time is the time between touching a key and touching the next key, whereas UU-time is the time between releasing a key and releasing the next key. UD-time is the time between releasing a key and touching the next key, whereas DU-time is the time between touching a key and releasing the next key. Fig. 9 illustrates an example of computing these five timing values and Table 8 lists the transformed five timing values for the example illustrated in Fig. 9.

Description	Press "Space"	Release "Space"
Ax	0.1217	-0.2050
Ay	-0.1618	-0.8613
Az	-0.0425	-0.0462

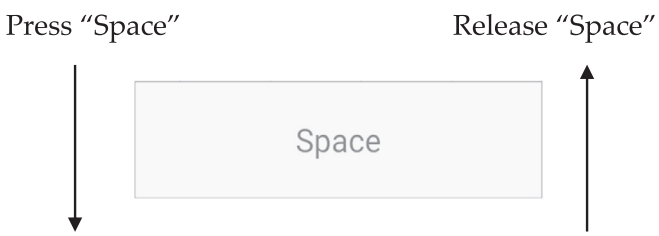


Fig. 6. Example of angular velocity variables while typing space.

Table 6
Description of variables for transformed rawdata.

Name	Description
Key_n	n^{th} pressed Key
TSD_n	Timestamp of a Key_n down
TSU_n	Timestamp of a Key_n up
Cx_n	X axis coordinate values collected from Key_n
Cy_n	Y axis coordinate values collected from Key_n
$X_{avg,\ n}$	Average of Cx_n
$Y_{avg,\ n}$	Average of Cy_n
$X_{max,n}$	Maximum of Cx_n
$Y_{max,n}$	Maximum of Cy_n
$X_{min,n}$	Minimum of Cx_n
$Y_{min,n}$	Minimum of Cy_n
AxD_n	Angular velocity of x axis when Key_n is down
AyD_n	Angular velocity of y axis when Key_n is down
AzD_n	Angular velocity of z axis when Key_n is down
AxU_n	Angular velocity of x axis when Key_n is up
AyU_n	Angular velocity of y axis when Key_n is up
AzU_n	Angular velocity of z axis when Key_n is up

Finally, we used the transformed three types of microbehavioral features for free-KDA: **A**-, **C**-, and **T**-based features. Table 9 lists all variables from the three groups used in our study.

4. Experimental settings

4.1. Data partition

As each participant provided 20 sets of English and Korean keystroke data, we used 10 sets for reference and the other 10 sets for testing. The average and standard deviation values of the keystroke length in each set is provided with the two-sided statistical test on the difference between the means of two populations in Table 10. Based on the probability value (p-value), it can be confirmed that there is no statistically significant difference between the keystroke lengths in the reference and test datasets. To evaluate user authentication performance in terms of the number of reference datasets, we varied the length of the reference dataset of the valid user from one to ten. For the test dataset, a total of 500 keystroke sets were used: 10 sets from the valid user and 490 sets from the impostors (10 test sets from the other participants, excluding the valid user).

4.2. Authentication model

4.2.1. Kolmogorov-Smirnov statistic and Cramér–von Mises criterion

One commonly used KDA method for freely typed text compares the keystroke feature distributions between two samples; if the two distributions are sufficiently similar, the two keystroke sets are considered to be typed by the same user, otherwise they are considered to be typed by different users. We used Kolmogorov-Smirnov (KS) statistic and CM criterion [23], which use non-parametric empirical distributions instead of parametric distributions, such as normal distribution to compare two keystroke feature distributions. Let D_{ref} and D_{tst} denote the keystroke datasets for reference and test, respectively. Let N_{ref} and N_{tst} denote the number of keystrokes in the reference and test datasets, respectively. Further, let X_i denote the i-th (i $\in$ {1,N}) data. Then, the cumulative distribution for the reference and test datasets, i.e. C_{ref} and C_{tst}, are defined as shown in Eq. (1–2):

$$C_{ref}(\alpha) = \frac{1}{N_{ref}} \sum_{i=1}^{N_{ref}} I(X_i \le \alpha), \tag{1}$$

$$C_{tst}(\alpha) = \frac{1}{N_{tst}} \sum_{i=1}^{N_{tst}} I(X_i \le \alpha), \tag{2}$$

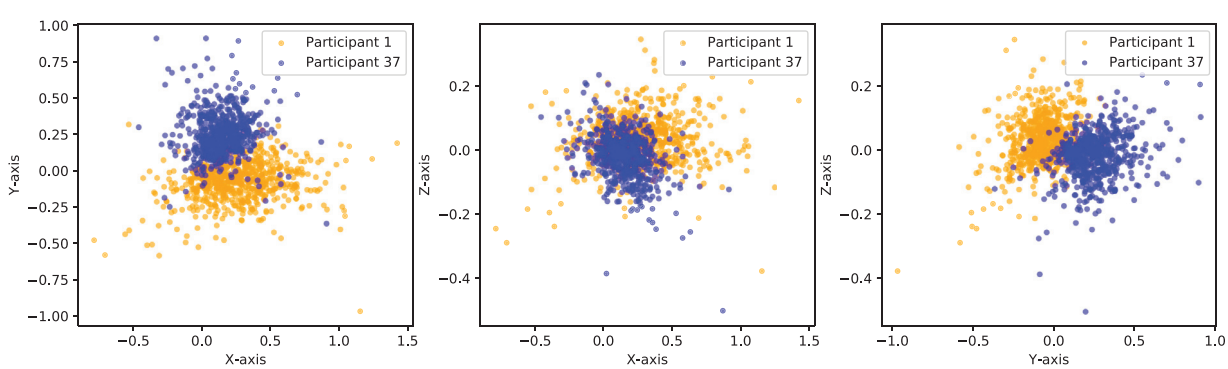


Fig. 7. Visualization of the three-axes acceleration value when the "space" key is pressed.

Table 7
Transformed data from rawdata1 and rawdata2.

Key	*TSD*	*AxD*	*AyD*	*AzD*	*TSU*	*AxU*	*AyU*	*AzU*	X_{avg}	Y_{avg}	X_{max}	Y_{max}	X_{min}	Y_{min}
a	0	0.62	-0.67	-0.18	250	-0.34	0.21	0.05	44.39	75.09	48.09	78.02	41.76	73.68
b	300	-0.14	0.12	-0.02	500	-0.05	0.09	0.02	40.92	96.04	41.96	97.11	39.88	94.97

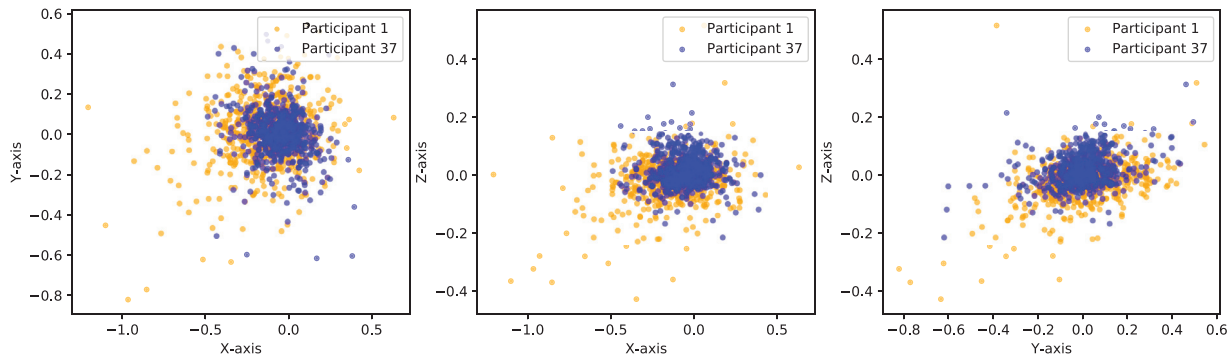


Fig. 8. Visualization of the three-axes acceleration value when the "space" key is released.

Table 8
Example of five basic keystroke times.

Key_1	Key_2	Duration Key_1	Duration Key_2	$DD_{1,2}$	$DU_{1,2}$	$UD_{1,2}$	$UU_{1,2}$
a	b	250	150	300	500	50	250

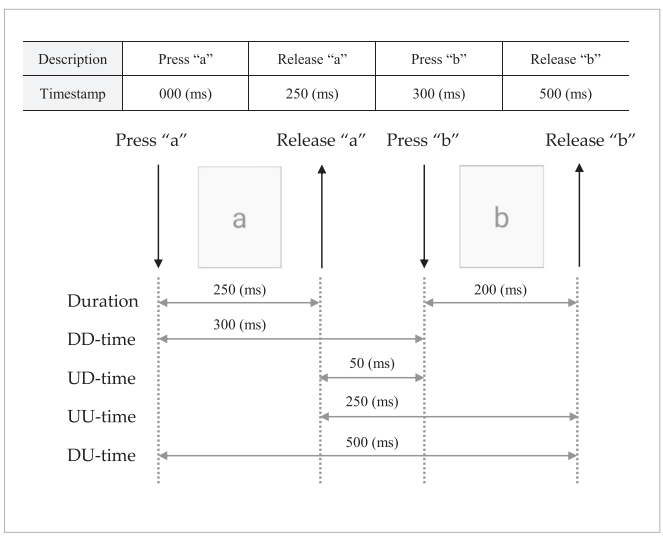


Fig. 9. Time capture moments and five basic keystroke times.

Table 9
Three groups of keystroke variables used in this study.

Name of variable group	Notation
A	*AxD, AyD, AzD, AxU, AyU, AzU*
C	X_{avg}, Y_{avg}, $X_{\max}$, $Y_{\max}$, $X_{\min}$, $Y_{\min}$
T	*Duration, DD, DU, UD, UU*

Table 10
Average and standard deviation of each keystroke dataset (numbers in parentheses denote the standard deviation and *p*-values for the two-sided test for the same mean values between two populations).

Language	Reference	Test	*p*-value
English	211.81 (36.39)	211.68 (39.30)	0.95
Korean	235.11 (42.23)	235.26 (42.84)	0.95

The I function is an indicator function, which returns one if the condition inside the parenthesis is satisfied and returns zero otherwise. Finally, the KS statistic was computed by Eq. (3):

$$KS_{D_{ref},D_{tst}} = sup|C_{ref}, C_{tst}| \tag{3}$$

Table 11
Unit of analysis for each microbehavioral feature.

Variable group	Notation
Unigraph	*AxD, AyD, AzD, AxU, AyU, AzU*, X_{avg}, Y_{avg}, $X_{\max}$, $Y_{\max}$, $X_{\min}$, $Y_{\min}$, *Duration*
Digraph	*DD, DU, UD, UU*

Fig. 10 (a) is a graphical representation of the KS statistic; moreover, it is the maximum difference between two cumulative density functions, $C_{ref}(X)$ and $C_{tst}(X)$. Hence, the more similar the typing behaviors between two users, the smaller the KS statistic. CM criterion was computed according to Eq. (4):

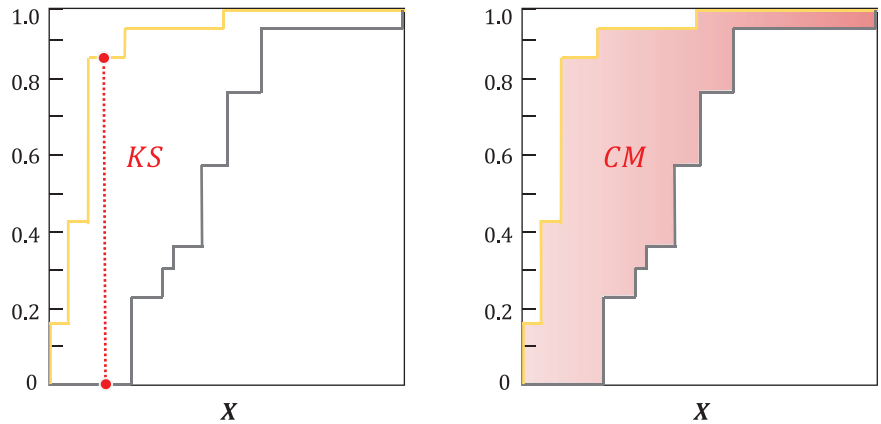
$$CM_{D_{ref},D_{tst}} = \int_{-\infty}^{\infty} [C_{ref}(X) - C_{tst}(X)]^2 dx \tag{4}$$

Fig. 10 (b) shows an illustrative example of the CM criterion with the same two empirical distributions as shown in Fig. 10 (a). The primary difference between the KS statistic and CM criterion is that the KS statistic uses the maximum difference between the two empirical distributions, whereas the CM criterion uses the cumulative difference between two distributions. In this study, KS statistic and CM criterion were computed for **A**-, **C**-, and **T**-based feature sets and they were averaged to compute the novelty score for user authentication.

4.2.2. TT and R measures

TT and R measures [20] compute the unigraph and digraph unit-level similarity based on microbehavioral features from two keystroke datasets, respectively, to determine whether they are typed by the same user. The unit of analysis for each microbehavioral feature is listed in Table 11. The main difference between the TT measure and R measure is that while the R measure primarily focuses on comparing the rank of microbehavioral features by computing the degree of disorder, the TT measure focuses on comparing the *p*-values from two sampled *t*-tests for the two microbehavioral feature samples.

Tables 12 and Table 13 are examples of unigraph and digraph key frequencies for two keystroke samples, respectively. In this study, we computed the TT and R measures based on the keys that were typed by both users at least twice. Therefore, b, h, and i are excluded in the unigraph key example. To compute the TT measure, two-sided unpaired *t*-tests were computed for every common key. The null and alternative hypotheses of the *t*-tests are shown in



(a) Kolmogorov-Smirnov statistic (b) Cramér–von Mises criterion

Fig. 10. Example of Kolmogorov-Smirnov statistic and Cramér–von Mises criterion.

Table 12
Number of unigraph keys typed in each sample.

Sample	space	b	s	d	delete	k	l	h	i
S1	10	5	4	3	20	5	4	3	0
S2	12	1	5	4	15	3	3	0	5

Table 13
Number of digraph keys typed in each sample.

Sample	delete/delete	s/delete	s/space	l/space
S1	10	3	3	3
S2	11	3	4	2

Table 14
Frequency, mean, and standard deviation of the unigraph key used in TT and R measures (numbers in parentheses denote the sample name).

Statistics	space	s	d	delete	k	l
Freq (S1)	10	4	3	20	5	4
Mean (S1)	150	220	230	270	280	290
Std (S1)	30	30	40	50	30	40
Freq (S2)	12	5	4	15	3	3
Mean (S2)	210	180	150	290	190	250
Std (S2)	40	35	20	60	20	30

Table 15
Frequency, mean, and standard deviation of the digraph key used in TT and R measures (numbers in parentheses denote the sample name).

Statistics	delete/delete	s/delete	s/space	l/space
Freq (S1)	10	3	3	3
Mean (S1)	350	420	390	440
Std (S1)	20	40	20	30
Freq (S2)	11	3	4	2
Mean (S2)	390	380	420	480
Std (S2)	25	30	30	40

Table 16
p-value based on each unigraph key (asterisks indicate that *p*-value is less than equal to significance level of 0.05).

Key	space	s	d	delete	k	l
p-value	0.0009*	0.1133	0.0167*	0.2902	0.0039*	0.2090

Table 17
p-value based on each digraph key (asterisks indicate that *p*-value is less than equal to significance level of 0.05).

Key	delete/delete	s/delete	s/space	l/space
p-value	0.0008*	0.2382	0.1978	0.2840

Eq. (5–6):

$$H_0 : \mu_{(S1,Keyi)} = \mu_{(S2,Keyi)} \tag{5}$$

$$H_1 : \mu_{(S1,Keyi)} \neq \mu_{(S2,Keyi)} \tag{6}$$

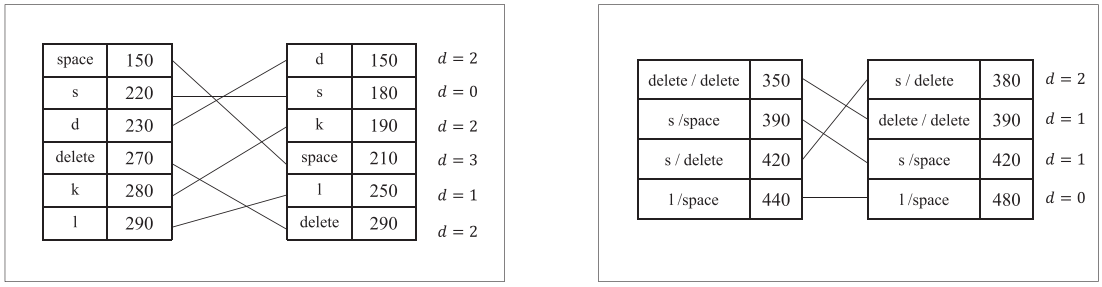
$$NT = \sum_{i=1}^{N} I(P_N \leq significance\ level), \tag{7}$$

$$TT = 1 - \frac{NT}{N} \tag{8}$$

The respective mean and standard deviation of duration for each unigraph and digraph key in Tables 12 and Table 13 are shown in Tables 14 and 15, respectively. The resulted *p*-values for the corresponding *t*-tests are listed in Tables 16 and 17. Based on the computed *p*-values and a user-defined significance level, we computed the number of keys that have lower *p*-values than the significance level (NT) as shown in Eq. (7). In this study, we set the significance level to 0.05. NT is three (space, d, and k) for the unigraph example and NT is one (delete/delete) for the digraph example. Finally, the TT measure was computed by Eq. (8):

In our example, TT is 0.5 for the unigraph example and 0.75 for the digraph example; the higher the TT value, the more is the similarity between two users.

The R measure computes the relative degree of disorder (d) between the key typing patterns of two participants. Each d from the common key is computed by the rank of mean value between two keystroke samples. In the previous example, each d is computed from the mean value of S1 and S2 in Tables 14 and Table 15, as shown in Fig. 11 (a) and (b). Therefore, the sum of the d values for the unigraph and digraph are 2+0+2+3+1+2=10 and 2+1+1+0=4, respectively. The maximum d based on the size of N common unigraphs or digraphs was computed by Eq. (9). In the unigraph and digraph examples, the maximum d values were 18 and 8, respectively. Finally, the R measure was computed as shown in Eq. (10). Hence, the R measure for the unigraph and digraph example were



(a) Uni-graph example (b) Di-graph example

Fig. 11. Example of degree of disorder.

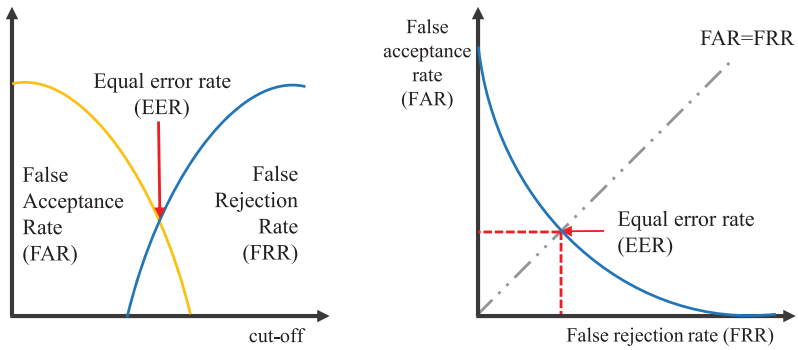


Fig. 12. FAR-FRR relationship with regard to the cut-off value and the meaning of equal error rate.

0.44 and 0.5, respectively.

$$Max.\ degree\ of\ disorder = \begin{cases} \frac{N^2}{2} & if\ \mathrm{n}\ is\ even, \\ \frac{N^2-1}{2} & if\ \mathrm{n}\ is\ odd, \end{cases} \tag{9}$$

$$R = 1 - \frac{Sum\ of\ degree\ of\ disorder}{Max.\ degree\ of\ disorder} \tag{10}$$

Next, TT plus R measure (TTPR) combines the TT and R measures by adding them, while the TT multiplied by R measure (TTMR) combines them by multiplication as shown in Eq. (11) and Eq. (12), respectively.

$$TTPR = TT + R \tag{11}$$

$$TTPR = TT \times R \tag{12}$$

In this study, TT, R, TTPR, and TTMR were computed for **A**-, **C**-, and **T**-based variables, similar to KS and CM. The arithmetic average of each variable set was finally used as a novelty score for user authentication.

4.2.3. Performance measure

EER was used as a performance measure in our study. Two types of errors occur in security systems: FRR and FAR. FRR is the proportion of falsely rejected trials of a valid user while FAR is the proportion of falsely accepted trials of an impostor. There is a trade-off between FRR and FAR, as shown in Fig. 12. When the authentication threshold is strictly set, a low FAR can be achieved at the cost of FRR. Conversely, a high FAR with a low FRR can be achieved when the threshold is loosely set. Hence, to evaluate the fundamental authentication ability, a threshold-independent performance measure is required. EER is the error rate at the point where the FRR and FAR become the same: the lower the EER, the better the authentication model. In this study, EER was computed based on the novelty scores obtained by 500 test keystroke sets (10 from a valid user and 490 from impostors).

5. Experimental results

The average EER and corresponding 95% confidence interval for each algorithm, using a single set from among the **A**-, **C**-, or **T**based feature sets, are provided in Table 18 for English keystrokes and in Table 19 for Korean keystrokes. In the case of KS and CM, which did not use the typed key information, the authentication performance was always ordered by **A, C**, and **T** for both English and Korean keystroke data. The EER is lower than 10% with **A**-based features for both languages although only one keystroke set was used for reference and it decreased to lower than 4% for English and 6% for Korean when the number of reference keystroke sets increased to 10. Conversely, the **T**-based measures were unable to achieve an EER lower than 10%, although 10 reference sets were used.

However, when the typed key information was utilized, the **T**-based features generally outperformed the **A**-based and **C**-based features for both languages. An interesting observation is that when the number of reference sets is very small, e.g., one, the **A**-based features resulted in lower EERs than the **T**-based features. However, the effect of an increased reference set is more significant for **T**-based features than **A**-based features; the EERs of TT and R measures with **T**-based features ranged from 1.75% to 4.04%, while those with **A**-based features ranged from 3.48% to 4.93%. Another observation with individual feature sets is that combining TT and R measures significantly improved the authentication performance, especially when the number of reference sets was large. When the number of reference sets was 10, TTPR and TTMR with **T**-based features resulted in an EER that was lower than 1% for three out of four cases (except the TTMR for Korean), which is between one-fourth and half of the EER values with either TT or R only.

The average EER and the corresponding 95% confidence interval for each algorithm, using a combination of two sets from among the **A**-, **C**-, and **T**-based features, are provided Table 20 for English keystrokes and Table 21 for Korean keystrokes. Obviously, the

Table 18
Average and 95% confidence interval of EER for each algorithm based on one feature set for English keystrokes (The values in bold denote the best performances for each algorithm-reference set size pair, while the asterisks indicate the best performance for the corresponding reference set size).

	KS			CM			TT			R			TTPR			TTMR		
No.Ref	A	C	T	A	C	T	A	C	T	A	C	T	A	C	T	A	C	T
1	**9.61**	11.14	17.46	**8.72**	15.34	17.57	**5.86**	9.64	8.38	**11.46**	25.16	11.69	**4.40***	11.01	5.63	**4.60**	14.50	6.44
	(± 3.68)	(± 2.85)	(± 2.99)	**(± 3.46)**	(± 2.67)	(± 3.12)	**(± 2.40)**	(± 1.86)	(± 2.19)	**(± 3.16)**	(± 2.90)	(± 2.44)	**(± 2.46)**	(± 2.07)	(± 1.81)	**(± 2.44)**	(± 2.09)	(± 1.91)
2	**5.79**	8.80	13.63	**4.74**	10.84	14.17	**3.61**	6.61	4.05	10.33	22.70	**8.00**	3.61	8.65	**2.76***	3.77	11.05	**2.93**
	(± 2.09)	(± 2.25)	(± 2.25)	**(± 2.08)**	(± 1.98)	(± 2.49)	**(± 1.97)**	(± 1.66)	(± 1.63)	(± 3.35)	(± 2.68)	**(± 1.84)**	(± 2.30)	(± 1.98)	**(± 1.34)**	(± 2.17)	(± 2.15)	**(± 1.23)**
3	**4.80**	8.59	12.11	**4.18**	10.25	12.75	3.66	6.14	**3.62**	8.07	20.07	**6.40**	3.00	6.86	**1.77***	3.03	8.92	**2.45**
	(± 1.97)	(± 2.56)	(± 1.80)	**(± 1.94)**	(± 2.09)	(± 2.08)	(± 1.85)	(± 1.70)	**(± 1.41)**	(± 3.12)	(± 2.84)	**(± 1.55)**	(± 2.05)	(± 1.85)	**(± 0.92)**	(± 2.05)	(± 2.04)	**(± 1.07)**
4	**4.76**	8.56	11.81	**3.29**	9.89	12.44	3.92	6.00	**2.54**	7.50	18.14	**5.34**	2.69	6.28	**1.47***	2.76	8.57	**1.49**
	(± 2.05)	(± 2.64)	(± 1.83)	**(± 1.44)**	(± 2.02)	(± 1.96)	(± 1.83)	(± 1.95)	**(± 1.10)**	(± 3.19)	(± 2.77)	**(± 1.53)**	(± 1.96)	(± 1.75)	**(± 0.87)**	(± 2.04)	(± 1.89)	**(± 0.82)**
5	**4.09**	8.28	11.54	**3.34**	9.41	12.19	3.32	5.72	**2.35**	6.36	16.92	**4.97**	2.57	5.56	**1.10***	2.59	7.83	**1.27**
	(± 1.92)	(± 2.57)	(± 1.77)	**(± 1.39)**	(± 1.92)	(± 1.97)	(± 1.87)	(± 1.83)	**(± 1.19)**	(± 3.08)	(± 2.68)	**(± 1.47)**	(± 2.01)	(± 1.76)	**(± 0.75)**	(± 2.02)	(± 2.05)	**(± 0.79)**
6	**4.17**	7.69	11.78	**3.20**	8.75	12.51	2.86	5.63	**2.74**	5.69	16.46	**4.66**	2.52	5.06	**1.15***	2.54	7.29	**1.16**
	(± 1.91)	(± 2.33)	(± 1.99)	**(± 1.37)**	(± 1.64)	(± 2.02)	(± 1.73)	(± 1.84)	**(± 1.40)**	(± 2.93)	(± 2.48)	**(± 1.55)**	(± 1.93)	(± 1.62)	**(± 0.74)**	(± 1.93)	(± 1.85)	**(± 0.74)**
7	**3.87**	7.98	11.61	**3.75**	8.57	11.97	3.02	5.71	**2.31**	5.34	15.14	**4.36**	2.03	4.87	**1.22***	2.11	6.04	**1.24**
	(± 1.78)	(± 2.29)	(± 2.09)	**(± 1.51)**	(± 1.87)	(± 1.96)	(± 1.66)	(± 1.85)	**(± 1.33)**	(± 2.69)	(± 2.58)	**(± 1.48)**	(± 1.54)	(± 1.39)	**(± 0.77)**	(± 1.58)	(± 1.50)	**(± 0.77)**
8	**4.03**	7.53	11.54	**3.20**	8.52	11.94	3.35	5.56	**2.24**	5.17	15.97	**4.17**	1.79	5.17	**1.04***	2.00	6.28	**1.08**
	(± 1.92)	(± 2.31)	(± 2.09)	**(± 1.38)**	(± 1.83)	(± 2.21)	(± 1.88)	(± 1.89)	**(± 1.20)**	(± 2.57)	(± 2.75)	**(± 1.39)**	(± 1.45)	(± 1.60)	**(± 0.73)**	(± 1.60)	(± 1.75)	**(± 0.75)**
9	**3.67**	7.27	11.48	**3.49**	8.35	11.74	3.39	5.75	**2.02**	5.31	15.16	**3.78**	2.00	4.98	**1.12***	2.04	6.23	**1.12***
	(± 1.82)	(± 2.10)	(± 1.96)	**(± 1.49)**	(± 1.84)	(± 2.04)	(± 1.82)	(± 1.59)	**(± 1.21)**	(± 2.62)	(± 2.72)	**(± 1.23)**	(± 1.57)	(± 1.39)	**(± 0.80)**	(± 1.63)	(± 1.59)	**(± 0.77)**
10	**3.45**	7.12	11.46	**3.10**	8.22	11.65	3.48	5.11	**2.46**	4.93	15.00	**3.87**	1.90	5.37	**0.99**	1.96	6.59	**0.96***
	(± 1.62)	(± 2.12)	(± 2.03)	**(± 1.35)**	(± 1.90)	(± 2.07)	(± 1.87)	(± 1.61)	**(± 1.31)**	(± 2.30)	(± 2.63)	**(± 1.22)**	(± 1.34)	(± 1.49)	**(± 0.73)**	(± 1.40)	(± 1.58)	**(± 0.68)**
Best	0	0	0	0	0	0	0	0	0	0	0	0	1	0	7.5	0	0	1.5

Table 19
Average EER and the corresponding 95% confidence interval for each algorithm based on one feature set for Korean keystrokes (The values in bold denote the best performances for each algorithm-reference set size pair, while the asterisks indicate the best performance for the corresponding reference set size).

	KS			CM			TT			R			TTPR			TTMR		
No.Ref	A	C	T	A	C	T	A	C	T	A	C	T	A	C	T	A	C	T
1	**9.03**	9.29	15.38	**9.04**	12.46	14.79	5.91	8.00	**4.53**	8.42	20.19	**8.14**	**3.16***	6.88	4.00	**3.28**	9.60	4.25
	(± 2.64)	(± 1.70)	(± 2.44)	**(± 2.91)**	(± 2.17)	(± 3.00)	(± 1.94)	(± 1.45)	**(± 1.38)**	(± 2.23)	(± 2.40)	**(± 2.02)**	**(± 1.37)**	(± 1.61)	(± 1.45)	**(± 1.36)**	(± 1.70)	(± 1.49)
2	**7.24**	8.48	13.43	**7.33**	10.98	13.42	**4.06**	4.75	4.64	6.40	16.21	**6.29**	1.71	4.77	**2.82***	**2.21**	5.98	2.53
	(± 2.27)	(± 1.74)	(± 2.50)	**(± 2.55)**	(± 1.83)	(± 2.92)	**(± 1.53)**	(± 1.46)	(± 1.65)	(± 1.71)	(± 2.88)	**(± 1.86)**	(± 0.92)	(± 1.36)	**(± 1.23)**	**(± 1.04)**	(± 1.52)	(± 1.1)
3	**6.51**	7.66	13.47	**6.43**	9.64	12.93	3.81	4.23	**2.29**	5.54	14.52	**5.24**	1.81	4.06	**1.57***	2.08	5.58	**1.61**
	(± 2.51)	(± 1.46)	(± 2.23)	**(± 2.45)**	(± 1.79)	(± 2.47)	(± 1.59)	(± 1.29)	**(± 1.17)**	(± 2.02)	(± 2.55)	**(± 1.81)**	(± 0.98)	(± 1.33)	**(± 0.88)**	(± 1.06)	(± 1.55)	**(± 0.91)**
4	**6.31**	7.16	12.70	**6.48**	9.35	12.38	3.58	3.59	**2.49**	4.99	12.19	**4.79**	1.46	3.09	**1.05***	1.33	4.27	**1.10**
	(± 2.46)	(± 1.38)	(± 2.34)	**(± 2.20)**	(± 1.64)	(± 2.43)	(± 1.60)	(± 1.18)	**(± 1.08)**	(± 1.50)	(± 2.21)	**(± 1.59)**	(± 0.88)	(± 1.27)	**(± 0.71)**	(± 0.84)	(± 1.55)	**(± 0.72)**
5	**5.71**	6.86	12.45	**6.36**	8.52	11.91	3.31	3.64	**1.98**	4.65	12.64	**3.88**	1.16	2.73	**1.04***	1.28	3.83	**1.21**
	(± 2.01)	(± 1.58)	(± 2.28)	**(± 2.18)**	(± 1.62)	(± 2.44)	(± 1.56)	(± 1.31)	**(± 1.00)**	(± 1.73)	(± 2.16)	**(± 1.41)**	(± 0.79)	(± 1.22)	**(± 0.76)**	(± 0.81)	(± 1.34)	**(± 0.82)**
6	**5.87**	6.59	11.88	**6.21**	8.37	11.11	3.69	3.82	**1.86**	4.38	11.47	**3.92**	1.23	2.63	**1.03***	1.07	3.33	**1.07**
	(± 2.04)	(± 1.49)	(± 2.18)	**(± 2.26)**	(± 1.55)	(± 2.28)	(± 1.50)	(± 1.30)	**(± 0.93)**	(± 1.66)	(± 2.36)	**(± 1.43)**	(± 0.81)	(± 1.17)	**(± 0.70)**	(± 0.75)	(± 1.32)	**(± 0.72)**
7	**6.25**	6.43	11.97	**6.25**	8.31	11.77	4.55	4.05	**2.04**	4.71	11.35	**3.88**	1.31	2.30	**0.88***	1.33	3.33	**1.03**
	(± 2.36)	(± 1.51)	(± 2.22)	**(± 2.29)**	(± 1.42)	(± 2.53)	(± 1.76)	(± 1.34)	**(± 0.97)**	(± 1.75)	(± 2.19)	**(± 1.42)**	(± 0.85)	(± 1.12)	**(± 0.62)**	(± 0.87)	(± 1.28)	**(± 0.68)**
8	**5.88**	6.20	11.80	**5.82**	8.15	11.52	3.83	3.88	**2.12**	4.29	11.19	**4.20**	1.36	2.34	**0.95***	1.25	3.16	**1.01**
	(± 2.41)	(± 1.56)	(± 2.17)	**(± 2.22)**	(± 1.53)	(± 2.41)	(± 1.76)	(± 1.29)	**(± 0.99)**	(± 1.79)	(± 2.22)	**(± 1.55)**	(± 0.85)	(± 1.14)	**(± 0.63)**	(± 0.81)	(± 1.28)	**(± 0.66)**
9	**5.46**	6.40	11.77	**5.71**	7.95	11.61	3.38	4.08	**1.71**	4.70	11.62	**4.09**	1.28	2.95	**0.47***	1.18	3.44	**0.89**
	(± 2.21)	(± 1.53)	(± 2.10)	**(± 2.22)**	(± 1.59)	(± 2.37)	(± 1.54)	(± 1.22)	**(± 0.87)**	(± 1.60)	(± 2.26)	**(± 1.40)**	(± 0.83)	(± 1.16)	**(± 0.21)**	(± 0.81)	(± 1.27)	**(± 0.58)**
10	**5.51**	6.14	12.17	**5.66**	7.69	11.34	3.82	4.21	**1.75**	4.69	11.53	**4.04**	1.34	2.75	**0.77***	1.35	3.57	**1.09**
	(± 2.21)	(± 1.52)	(± 2.06)	**(± 2.26)**	(± 1.56)	(± 2.41)	(± 1.72)	(± 1.24)	**(± 0.92)**	(± 1.73)	(± 2.34)	**(± 1.48)**	(± 0.87)	(± 1.12)	**(± 0.55)**	(± 0.89)	(± 1.26)	**(± 0.72)**
Best	0	0	0	0	0	0	0	0	0	0	0	0	2	0	8	0	0	0

Table 20
Average EER and the corresponding 95% confidence interval for each algorithm based on two feature sets for English keystrokes (The values in bold denote the best performances for each algorithm-reference set size pair, while the asterisks indicate the best performance for the corresponding reference set size).

	KS			CM			TT			R			TTPR			TTMR		
No.Ref	AT	AC	TC	AT	AC	TC	AT	AC	TC	AT	AC	TC	AT	AC	TC	AT	AC	TC
1	**3.30**	6.31	6.96	**4.48**	7.12	9.25	**2.19**	2.24	3.38	11.22	**4.46**	8.21	2.33	**1.20***	1.74	3.49	**1.33**	2.64
	(± 1.62)	(± 2.40)	(± 2.45)	**(± 1.96)**	(± 2.34)	(± 2.50)	**(± 1.22)**	(± 1.22)	(± 1.22)	(± 2.31)	**(± 1.76)**	(± 1.98)	(± 1.23)	**(± 0.80)**	(± 0.93)	(± 1.54)	**(± 0.89)**	(± 1.09)
2	**1.90**	3.38	5.32	**2.38**	4.02	5.79	0.72	**0.60**	1.16	8.25	**3.01**	5.53	1.29	**0.72***	0.92	1.97	**0.92**	1.18
	(± 1.27)	(± 1.31)	(± 1.82)	**(± 1.17)**	(± 1.45)	(± 1.70)	(± 0.63)	**(± 0.76)**	(± 0.78)	(± 2.15)	**(± 1.45)**	(± 1.52)	(± 0.98)	**(± 0.87)**	(± 0.74)	(± 1.21)	**(± 0.92)**	(± 0.82)
3	**2.07**	2.19	3.61	**1.78**	2.77	4.98	0.74	**0.68**	0.99	6.24	**1.95**	3.98	0.97	**0.59***	0.70	1.28	**0.60**	0.75
	(± 1.77)	(± 1.02)	(± 1.44)	**(± 1.26)**	(± 1.12)	(± 1.53)	(± 0.82)	**(± 0.61)**	(± 0.74)	(± 2.21)	**(± 1.15)**	(± 1.47)	(± 0.92)	**(± 0.62)**	(± 0.68)	(± 1.05)	**(± 0.62)**	(± 0.68)
4	**1.94**	2.09	2.98	**1.81**	2.89	5.09	0.77	0.93	**0.73**	5.42	**1.37**	2.21	0.86	**0.50**	0.59	0.96	**0.47***	0.64
	(± 1.43)	(± 0.98)	(± 1.48)	**(± 1.15)**	(± 1.11)	(± 1.58)	(± 0.89)	(± 0.73)	**(± 0.65)**	(± 2.03)	**(± 0.87)**	(± 0.98)	(± 0.94)	**(± 0.56)**	(± 0.60)	(± 0.96)	**(± 0.54)**	(± 0.63)
5	**1.76**	2.32	3.54	**2.01**	3.03	5.46	**0.50**	0.83	0.70	4.58	**1.51**	1.90	0.71	0.43	**0.33***	1.00	**0.43**	0.47
	(± 1.40)	(± 1.05)	(± 1.63)	**(± 1.14)**	(± 1.15)	(± 1.68)	**(± 0.56)**	(± 0.74)	(± 0.67)	(± 2.02)	**(± 1.02)**	(± 0.93)	(± 0.87)	(± 0.52)	**(± 0.41)**	(± 1.00)	**(± 0.53)**	(± 0.51)
6	**1.71**	2.43	3.39	**1.90**	2.94	5.13	**0.51**	0.94	0.77	3.85	**1.36**	2.17	0.71	0.42	**0.34***	0.87	**0.39**	0.38
	(± 1.33)	(± 1.10)	(± 1.66)	**(± 1.01)**	(± 1.28)	(± 1.63)	**(± 0.52)**	(± 0.74)	(± 0.70)	(± 1.77)	**(± 0.88)**	(± 1.20)	(± 0.82)	(± 0.50)	**(± 0.42)**	(± 0.94)	**(± 0.44)**	(± 0.42)
7	**1.80**	2.17	3.21	**2.00**	2.78	5.13	0.53	0.80	**0.49**	3.48	**1.19**	2.24	0.50	**0.27**	0.28	0.77	**0.22***	0.30
	(± 1.39)	(± 1.07)	(± 1.65)	**(± 1.15)**	(± 1.24)	(± 1.50)	(± 0.51)	(± 0.66)	**(± 0.54)**	(± 1.58)	**(± 0.78)**	(± 1.31)	(± 0.60)	**(± 0.41)**	(± 0.41)	(± 0.90)	**(± 0.31)**	(± 0.41)
8	**1.69**	2.04	3.20	**1.81**	2.62	5.33	0.57	0.69	**0.52**	3.62	**0.98**	1.89	0.49	**0.23**	0.29	0.66	**0.20***	0.31
	(± 1.42)	(± 1.03)	(± 1.66)	**(± 1.12)**	(± 1.17)	(± 1.51)	(± 0.56)	(± 0.65)	**(± 0.56)**	(± 1.59)	**(± 0.85)**	(± 1.23)	(± 0.77)	**(± 0.38)**	(± 0.41)	(± 0.86)	**(± 0.32)**	(± 0.41)
9	**1.52**	2.11	3.23	**1.87**	3.04	5.11	0.57	0.56	**0.47**	3.57	**0.98**	1.72	0.32	**0.10**	0.28	0.53	**0.08***	0.32
	(± 1.14)	(± 1.05)	(± 1.64)	**(± 1.02)**	(± 1.28)	(± 1.41)	(± 0.54)	(± 0.57)	**(± 0.53)**	(± 1.64)	**(± 0.87)**	(± 1.05)	(± 0.43)	**(± 0.10)**	(± 0.40)	(± 0.60)	**(± 0.09)**	(± 0.41)
10	**1.38**	1.89	3.26	**1.82**	2.83	5.01	0.70	0.50	**0.49**	3.45	**0.70**	1.50	0.30	**0.10**	0.27	0.52	**0.09***	0.30
	(± 1.07)	(± 1.00)	(± 1.64)	**(± 1.12)**	(± 1.28)	(± 1.45)	(± 0.62)	(± 0.51)	**(± 0.53)**	(± 1.60)	**(± 0.60)**	(± 1.01)	(± 0.41)	**(± 0.11)**	(± 0.40)	(± 0.57)	**(± 0.09)**	(± 0.41)
Best	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	0	5	0

Table 21
Average EER and the corresponding 95% confidence interval for each algorithm based on two feature sets for Korean keystrokes (The values in bold denote the best performances for each algorithm-reference set size pair, while the asterisks indicate the best performance for the corresponding reference set size).

	KS			CM			TT			R			TTPR			TTMR		
No.Ref	AT	AC	TC	AT	AC	TC	AT	AC	TC	AT	AC	TC	AT	AC	TC	AT	AC	TC
1	**3.54**	5.20	4.74	**4.22**	5.47	6.77	1.74	1.49	**1.28**	6.30	**2.75**	5.25	0.93	0.69	**0.64**	1.35	**0.62***	0.79
	(± 1.50)	(± 1.90)	(± 1.50)	**(± 1.74)**	(± 2.08)	(± 1.80)	(± 0.97)	(± 1.03)	**(± 0.72)**	(± 1.62)	**(± 1.52)**	(± 1.62)	(± 0.72)	(± 0.60)	**(± 0.49)**	(± 0.85)	**(± 0.56)**	(± 0.52)
2	**2.13**	3.60	3.84	**2.83**	4.77	7.51	**0.68**	1.25	0.87	3.44	**1.16**	2.36	0.29	**0.09**	0.30	0.43	**0.08***	0.19
	(± 1.17)	(± 1.71)	(± 1.44)	**(± 1.37)**	(± 1.92)	(± 2.21)	**(± 0.63)**	(± 0.85)	(± 0.55)	(± 1.35)	**(± 0.79)**	(± 1.08)	(± 0.32)	**(± 0.08)**	(± 0.31)	(± 0.43)	**(± 0.06)**	(± 0.10)
3	**2.07**	3.25	3.20	**2.65**	3.95	5.88	0.90	0.64	**0.51**	2.86	**0.61**	1.81	0.28	**0.04**	0.22	0.31	**0.03***	0.26
	(± 1.05)	(± 1.55)	(± 1.17)	**(± 1.28)**	(± 1.68)	(± 1.78)	(± 0.76)	(± 0.56)	**(± 0.52)**	(± 1.16)	**(± 0.53)**	(± 0.93)	(± 0.40)	**(± 0.04)**	(± 0.31)	(± 0.40)	**(± 0.03)**	(± 0.33)
4	**1.59**	3.39	2.73	**2.59**	3.45	5.58	0.51	0.35	**0.21**	2.28	**0.83**	1.48	0.39	**0.01***	0.06	0.43	**0.01***	0.07
	(± 0.91)	(± 1.40)	(± 1.07)	**(± 1.24)**	(± 1.50)	(± 1.77)	(± 0.53)	(± 0.41)	**(± 0.13)**	(± 1.02)	**(± 0.66)**	(± 0.79)	(± 0.51)	**(± 0.01)**	(± 0.05)	(± 0.55)	**(± 0.01)**	(± 0.06)
5	**1.17**	2.65	2.72	**2.42**	3.13	5.18	0.64	0.48	**0.10**	1.83	**0.59**	1.28	0.20	**0.01***	0.02	0.23	**0.01***	0.02
	(± 0.72)	(± 1.47)	(± 1.08)	**(± 1.20)**	(± 1.36)	(± 1.61)	(± 0.62)	(± 0.54)	**(± 0.06)**	(± 0.92)	**(± 0.54)**	(± 0.72)	(± 0.33)	**(± 0.01)**	(± 0.02)	(± 0.34)	**(± 0.01)**	(± 0.02)
6	**1.34**	2.38	2.57	**2.48**	3.14	5.38	0.54	0.32	**0.10**	1.48	**0.52**	1.04	0.07	**0.02***	0.04	0.09	**0.02***	0.05
	(± 0.80)	(± 1.18)	(± 1.05)	**(± 1.11)**	(± 1.61)	(± 1.68)	(± 0.52)	(± 0.40)	**(± 0.07)**	(± 0.78)	**(± 0.50)**	(± 0.72)	(± 0.08)	**(± 0.02)**	(± 0.03)	(± 0.08)	**(± 0.02)**	(± 0.05)
7	**1.31**	2.48	2.56	**2.69**	3.07	5.37	0.73	0.31	**0.12**	1.60	**0.57**	1.05	0.19	**0.01***	0.04	0.19	**0.01***	0.05
	(± 0.80)	(± 1.21)	(± 1.06)	**(± 1.27)**	(± 1.68)	(± 1.75)	(± 0.66)	(± 0.37)	**(± 0.07)**	(± 0.85)	**(± 0.51)**	(± 0.69)	(± 0.33)	**(± 0.01)**	(± 0.04)	(± 0.32)	**(± 0.01)**	(± 0.05)
8	**1.34**	2.17	2.74	**2.31**	3.40	5.60	0.71	0.46	**0.09**	1.89	**0.40**	0.95	0.10	**0.01***	0.04	0.23	**0.01***	0.05
	(± 0.80)	(± 1.13)	(± 1.11)	**(± 1.19)**	(± 1.74)	(± 1.78)	(± 0.66)	(± 0.55)	**(± 0.08)**	(± 0.95)	**(± 0.35)**	(± 0.64)	(± 0.11)	**(± 0.01)**	(± 0.06)	(± 0.32)	**(± 0.01)**	(± 0.08)
9	**1.28**	2.00	2.91	**2.35**	3.00	5.73	0.68	0.43	**0.10**	1.66	**0.53**	0.98	0.08	**0.01***	0.03	0.10	**0.01***	0.04
	(± 0.77)	(± 1.10)	(± 1.11)	**(± 1.23)**	(± 1.06)	(± 1.86)	(± 0.62)	(± 0.54)	**(± 0.09)**	(± 0.90)	**(± 0.46)**	(± 0.65)	(± 0.07)	**(± 0.01)**	(± 0.05)	(± 0.09)	**(± 0.01)**	(± 0.06)
10	**1.23**	1.76	2.76	**2.21**	2.99	5.54	0.71	0.43	**0.11**	1.91	**0.35**	0.93	0.09	**0.01***	0.02	0.12	**0.01***	0.03
	(± 0.75)	(± 1.05)	(± 1.09)	**(± 1.07)**	(± 1.57)	(± 1.83)	(± 0.60)	(± 0.53)	**(± 0.09)**	(± 0.96)	**(± 0.32)**	(± 0.61)	(± 0.09)	**(± 0.01)**	(± 0.03)	(± 0.12)	**(± 0.01)**	(± 0.04)
Best	0	0	0	0	0	0	0	0	0	0	0	0	0	3.5	0	0	6.5	0

Table 22
Average EER and the corresponding 95% confidence interval for each algorithm based on three feature sets for English keystrokes (The values in bold denote the best performances for the corresponding reference set size).

No.REF	KS ACT	CM ACT	TT ACT	R ACT	TTPR ACT	TTMR ACT
1	3.02(± 1.52)	3.66(± 1.80)	0.66(± 0.51)	3.62(± 1.47)	**0.45(± 0.44)**	0.69(± 0.58)
2	1.72(± 1.10)	1.60(± 0.89)	**0.04(± 0.04)**	2.12(± 1.21)	0.23(± 0.32)	0.30(± 0.34)
3	1.19(± 0.84)	1.24(± 0.79)	**0.07(± 0.07)**	1.92(± 1.20)	0.23(± 0.39)	0.39(± 0.51)
4	1.23(± 0.88)	1.05(± 0.69)	0.23(± 0.36)	1.11(± 0.93)	**0.22(± 0.40)**	0.23(± 0.40)
5	1.21(± 1.02)	1.00(± 0.67)	**0.21(± 0.40)**	0.87(± 0.75)	**0.21(± 0.40)**	0.22(± 0.40)
6	1.09(± 0.85)	1.39(± 0.84)	**0.17(± 0.31)**	0.91(± 0.88)	0.21(± 0.40)	0.22(± 0.40)
7	1.29(± 1.08)	0.94(± 0.68)	**0.18(± 0.32)**	0.89(± 0.95)	0.21(± 0.40)	0.21(± 0.38)
8	1.27(± 1.07)	0.95(± 0.68)	**0.06(± 0.09)**	0.74(± 0.89)	0.20(± 0.40)	0.19(± 0.37)
9	1.24(± 1.03)	1.04(± 0.72)	**0.06(± 0.06)**	0.61(± 0.63)	0.16(± 0.32)	0.18(± 0.34)
10	1.24(± 1.02)	1.12(± 0.77)	**0.07(± 0.08)**	0.51(± 0.55)	0.18(± 0.34)	0.17(± 0.34)
Best	0	0	6.5	0	2.5	0

Table 23
Average EER and the corresponding 95% confidence interval for each algorithm based on three feature sets for Korean keystrokes (The values in bold denote the best performances for the corresponding reference set size).

No.REF	KS ACT	CM ACT	TT ACT	R ACT	TTPR ACT	TTMR ACT
1	2.39(± 1.23)	3.24(± 1.56)	0.42(± 0.37)	1.91(± 1.01)	**0.08(± 0.05)**	0.13(± 0.10)
2	1.71(± 1.01)	2.03(± 1.37)	0.23(± 0.32)	0.69(± 0.64)	**0.02(± 0.03)**	**0.02(± 0.02)**
3	1.42(± 0.85)	1.66(± 1.07)	0.10(± 0.08)	0.30(± 0.36)	**0.02(± 0.03)**	**0.02(± 0.03)**
4	0.84(± 0.64)	1.43(± 0.99)	0.04(± 0.04)	0.12(± 0.09)	**0.00(± 0.00)**	**0.00(± 0.00)**
5	0.67(± 0.57)	0.98(± 0.87)	0.01(± 0.01)	0.12(± 0.10)	**0.00(± 0.00)**	**0.00(± 0.00)**
6	0.50(± 0.43)	1.09(± 0.85)	0.02(± 0.02)	0.08(± 0.09)	**0.00(± 0.00)**	**0.00(± 0.00)**
7	0.67(± 0.56)	1.12(± 0.74)	0.02(± 0.03)	0.20(± 0.32)	**0.00(± 0.00)**	**0.00(± 0.00)**
8	0.69(± 0.57)	1.11(± 0.74)	0.01(± 0.01)	0.20(± 0.31)	**0.00(± 0.00)**	**0.00(± 0.00)**
9	0.48(± 0.42)	1.05(± 0.67)	0.01(± 0.01)	0.20(± 0.31)	**0.00(± 0.00)**	**0.00(± 0.00)**
10	0.47(± 0.42)	1.29(± 0.80)	0.01(± 0.02)	0.04(± 0.04)	**0.00(± 0.00)**	**0.00(± 0.00)**
Best	0	0	0	0	5.5	4.5

authentication performance was improved when heterogeneous feature sets were fused; any two feature sets resulted in lower EER than the best single feature set for all numbers of reference keystroke sets. For both KS and CM, which do not utilize typed key information, the fusion of **A**- and **T**-based features (AT) always yielded the best EER and it steadily decreased as the number of reference keystroke sets was increased. When 10 sets of reference keystroke sets were used, the EER was between 1.23% and 2.21%. When the typed key information was considered, the fusion of **A**- and **C**-based features generally performed better than other combinations, except the TT method. Almost 1% of EER can be achieved with only one reference keystroke set with TTMR method for English; moreover, it was even lower than 0.7% for Korean. When the number of reference keystroke sets was increased to 10, the EER decreased to 0.09% for English and 0.01% for Korean; when users typed Korean, 1 out of 10,000 trials were falsely accepted or rejected, which can be considered as a very strong authentication method in practice.

The average EER and the corresponding 95% confidence interval for each algorithm, using the proposed FACT model that used a combination of all three **A**-, **C**-, and **T**-based feature sets, are provided Table 22 for English keystrokes and Table 23 for Korean keystrokes. Similar to the previous results, it was determined that the fusion of more heterogeneous features clearly improved the authentication performance; the EER of each authentication algorithm with the fusion of three features sets was always lower than all single feature sets and all combinations of two feature sets under the same number of reference keystroke sets and language setting. The best authentication model differed according to the language; TT resulted in the lowest EER for most cases of English while TTPR and TTMR resulted in the lowest EER for all cases, excluding one case (TTMR with one set of reference keystrokes). An interesting observation is that the EER with Korean was generally lower than that with English under the other settings are identical. This is primarily because the participants recruited in our study were all Koreans; consequently, they were more familiar with the Korean alphabet (Hangul) than the English alphabet; therefore, the typing habits were more distinctive and unique for each participant when he/she typed the script in Korean rather than English, which was also identified in a previous study [45]. Although the EER with English was higher than that of Korean, it was still sufficiently strong to be implemented in a real authentication system. It can protect more than 99.5% of impostor trials with less than 0.5% of FRR with only one set of reference keystrokes; further, it was strengthened to protect 99.93% of impostor trials with less than 0.07% of FRR when ten sets of reference keystrokes were available. The results are more surprising for Korean; all impostor trials could be detected while all trials of a valid user were granted access (perfect authentication becomes possible) when more than four sets of reference keystroke sets were provided.

Tables 24 and 25 present the average EERs and corresponding 95% confidence intervals of all the authentication methods according to different numbers of reference keystroke sizes and different keystroke feature sets. Based on these tables, irrespective of the language used in the experiment, combining heterogeneous keystroke feature sets always helped in improving the authentication performances. All combinations of two feature sets resulted in lower EERs, with significant margins, than those for the best single feature set. For all the reference keystroke sizes, the proposed FACT model that integrates three different keystroke feature sets, even lowered the EERs and resulted in the smallest variations. These performance improvements of the FACT model over all other feature combinations are statistically supported at a significance level of 0.05. One of the primary practical burdens of the KDA system

Table 24
Average EER and the corresponding 95% confidence interval based on all authentication algorithms for each reference set size with English keystrokes (The values in bold denote the best performance among the feature set combinations, single asterisks indicate that the average EER obtained via the ACT feature is less than that obtained via all other feature methods at significance level of 0.05).

No. REF	A	C	T	AT	AC	TC	ACT
1	7.44 ± 1.22	14.47 ± 1.13	11.2 ± 1.12	4.50 ± 0.76	3.78 ± 0.72	5.36 ± 0.79	**2.02 ± 0.50***
2	5.31 ± 0.98	11.44 ± 1.04	7.59 ± 0.92	2.75 ± 0.60	2.11 ± 0.48	3.32 ± 0.58	**1.00 ± 0.32***
3	4.46 ± 0.89	10.14 ± 1.02	6.52 ± 0.78	2.18 ± 0.61	1.46 ± 0.37	2.50 ± 0.50	**0.84 ± 0.30***
4	4.15 ± 0.88	9.57 ± 0.99	5.85 ± 0.77	1.96 ± 0.54	1.37 ± 0.35	2.04 ± 0.46	**0.68 ± 0.26***
5	3.71 ± 0.85	8.95 ± 0.97	5.57 ± 0.76	1.76 ± 0.52	1.43 ± 0.36	2.07 ± 0.48	**0.62 ± 0.26***
6	3.50 ± 0.81	8.48 ± 0.90	5.66 ± 0.81	1.59 ± 0.47	1.41 ± 0.37	2.03 ± 0.49	**0.66 ± 0.27***
7	3.35 ± 0.74	8.05 ± 0.87	5.45 ± 0.79	1.51 ± 0.45	1.24 ± 0.34	1.94 ± 0.49	**0.62 ± 0.28***
8	3.26 ± 0.74	8.17 ± 0.92	5.34 ± 0.79	1.47 ± 0.46	1.13 ± 0.33	1.92 ± 0.49	**0.57 ± 0.27***
9	3.32 ± 0.75	7.96 ± 0.85	5.21 ± 0.77	1.40 ± 0.41	1.14 ± 0.35	1.85 ± 0.46	**0.55 ± 0.25***
10	3.14 ± 0.67	7.90 ± 0.86	5.23 ± 0.78	1.36 ± 0.41	1.02 ± 0.31	1.81 ± 0.45	**0.55 ± 0.24***
Best	0	0	0	0	0	0	10

Table 25
Average EER and the corresponding 95% confidence interval based on all authentication algorithms for each reference set size with Korean keystrokes (The values in bold denote the best performance among the feature set combinations, single asterisks indicate that the average EER obtained via the ACT feature is less than that obtained via all other feature methods at significance level of 0.05).

No. REF	A	C	T	AT	AC	TC	ACT
1	6.47 ± 0.90	11.07 ± 0.89	8.51 ± 0.99	3.01 ± 0.56	2.70 ± 0.61	3.24 ± 0.57	**1.36 ± 0.39***
2	4.82 ± 0.75	8.53 ± 0.87	7.19 ± 0.94	1.63 ± 0.42	1.82 ± 0.50	2.51 ± 0.56	**0.78 ± 0.32***
3	4.36 ± 0.78	7.62 ± 0.79	6.19 ± 0.89	1.51 ± 0.38	1.42 ± 0.43	1.98 ± 0.46	**0.58 ± 0.25***
4	4.02 ± 0.72	6.61 ± 0.73	5.75 ± 0.86	1.30 ± 0.35	1.34 ± 0.39	1.69 ± 0.42	**0.41 ± 0.20***
5	3.74 ± 0.68	6.37 ± 0.73	5.41 ± 0.84	1.08 ± 0.31	1.14 ± 0.38	1.55 ± 0.40	**0.30 ± 0.17***
6	3.74 ± 0.68	6.04 ± 0.72	5.15 ± 0.80	1.00 ± 0.29	1.07 ± 0.36	1.53 ± 0.41	**0.28 ± 0.16***
7	4.07 ± 0.73	5.96 ± 0.70	5.26 ± 0.83	1.12 ± 0.32	1.07 ± 0.38	1.53 ± 0.42	**0.34 ± 0.16***
8	3.74 ± 0.72	5.82 ± 0.70	5.27 ± 0.82	1.10 ± 0.32	1.08 ± 0.38	1.58 ± 0.42	**0.33 ± 0.17***
9	3.62 ± 0.68	6.08 ± 0.70	5.09 ± 0.80	1.02 ± 0.31	1.00 ± 0.36	1.63 ± 0.44	**0.29 ± 0.14***
10	3.73 ± 0.71	5.98 ± 0.71	5.19 ± 0.82	1.05 ± 0.29	0.92 ± 0.35	1.57 ± 0.42	**0.30 ± 0.16***
Best	0	0	0	0	0	0	10

in a real-time authentication system is that it typically requires users to provide a significantly large amount of keystroke data at the initial stage to build an accurate authentication model. When more keystrokes are provided, a more robust authentication model can be obtained. However, from the perspective of system users, it would be inconvenient to type a large number of keystrokes at once. Because the proposed FACT model can achieve an EER as low as 1% with only one reference keystroke set, the inconvenience mentioned earlier can be significantly reduced. Once a user is registered in the system, they are required to provide only one set of freely typed keystrokes to initiate their authentication model. Whenever they type more text later, these additional keystrokes can be added to the reference keystroke set to strengthen the current authentication model.

In conclusion, our proposed FACT model based on 17 microbehavioral features consisting of not only typing pattern but also hand size, finger length, fingertip size, and muscle flexibility can produce an outstanding user authentication performance in mobile devices for free-KDA. In addition, we have publicized our collected keystroke dataset so that anyone who wants to conduct a KDA study or develop a KDA-related mobile service can use the dataset without any restrictions. As real data acquisition is one of the main obstacles in KDA-related studies, we expect that the developed FACT method as well as the released real keystroke dataset can contribute to the research and industrial communities.

6. Conclusion

In this study, we proposed a new user authentication model for mobile devices named FACT. In the proposed model, we integrated three types of features: **A**-, **C**-, and **T**-based features. We collected these features from freely typed keystroke data using an Android-based keystroke data collector application. From the raw data, we extracted 17 microbehavioral features relating to not only the typing pattern but also hand size, finger length, fingertip size, and muscle flexibility. To verify the proposed FACT model, we collected keystroke data from 50 participants who typed 20 predefined freely typed English and Korean texts. We used KS, CM, TT, R, TTPR, and TTMR in our authentication algorithm and the data of only one valid user was used as the reference set. Further, the experiment was performed for all possible feature set combinations.

The experimental results can be summarized as follows. First, using a single feature set, the TT and R measures, which utilized the typed key information, outperformed KS and CM, which do not utilize the typed key information. These results imply that **A**-, **C**-, and **T**-based microbehavior features using typed key information have a positive effect in characterizing the distinctive typing behavior of a user. Second, the fusion of heterogeneous feature sets generally helps to improve the authentication performance. All combinations of two feature sets resulted in the lowest EERs than that of the best single feature set. Similarly, the proposed FACT model, which fused all three feature sets, yielded much lower EERs than any two feature set combinations under all language and reference set size settings. When only one reference keystroke set was used, FACT resulted in an EER of up to 0.45% and 0.08% in English and Korean, respectively. When 10 reference keystroke sets were used, the EER was significantly reduced up to 0.07% and 0% in English and Korean, respectively. Especially, in the case of Korean, FACT yielded an average EER of 0%, although only four reference keystroke sets were used. Thus, it can com-

pletely discriminate the valid user from potential impostors when a certain number of keystroke sets are provided (four sets in our experiment).

Despite the favorable experimental results, there are certain limitations in the current study, which has led us in the future research directions. First, all the participants recruited were native Korean speakers for the keystroke data collection. It was identified that the average EER with Korean keystrokes was generally lower than that with English keystrokes. Hence, there is a requirement for a future study on a systematic analysis of the effect of language on KDA. Second, we used the fusion of three heterogeneous data collected from three sensors, i.e., accelerometer, coordinate of touchscreen, and touching time. As the experimental results showed that integrating heterogeneous features consistently improved user authentication performance, we should explore other types of built-in sensors in smartphone that can generate data while typing text for better authentication performance. Concerning the experimental settings, we conducted a carefully designed experiment to verify the fusion of different keystroke feature sets. To this end, we must control other factors that would influence the authentication performance, such as typing hands and typing positions. Because the KDA performance was outstanding under this well-controlled environment, we should extend the research to checking whether the FACT model would still maintain the reported authentication performance under less-controlled typing circumstances. Finally, in the current experiment, the proposed FACT uses one text set for free-KDA. Hence, whether high performance would still be maintained when freely typed keystrokes are used for continuous user authentication should be studied [46].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

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